



Smart Traffic Violation Management System

MERN Stack Based Smart System

Names:

Vivan Shetty- 10637

Cian Fonseca-10909

Soham More-10622

Computer Science Department

Course

B.Tech Computer Engineering

FSD Mini Project

Prof.Incharge- Sohan Agate

Academic Year 2026

The Problem with Manual Systems

Inefficiency

Manual violation logging is slow, error-prone, and resource-intensive — creating massive backlogs.

No Centralization

Violation data is siloed across departments with no unified tracking or cross-referencing.

Repeat Offenders

No mechanism exists to flag or track individuals with a history of multiple violations.

Zero Analytics

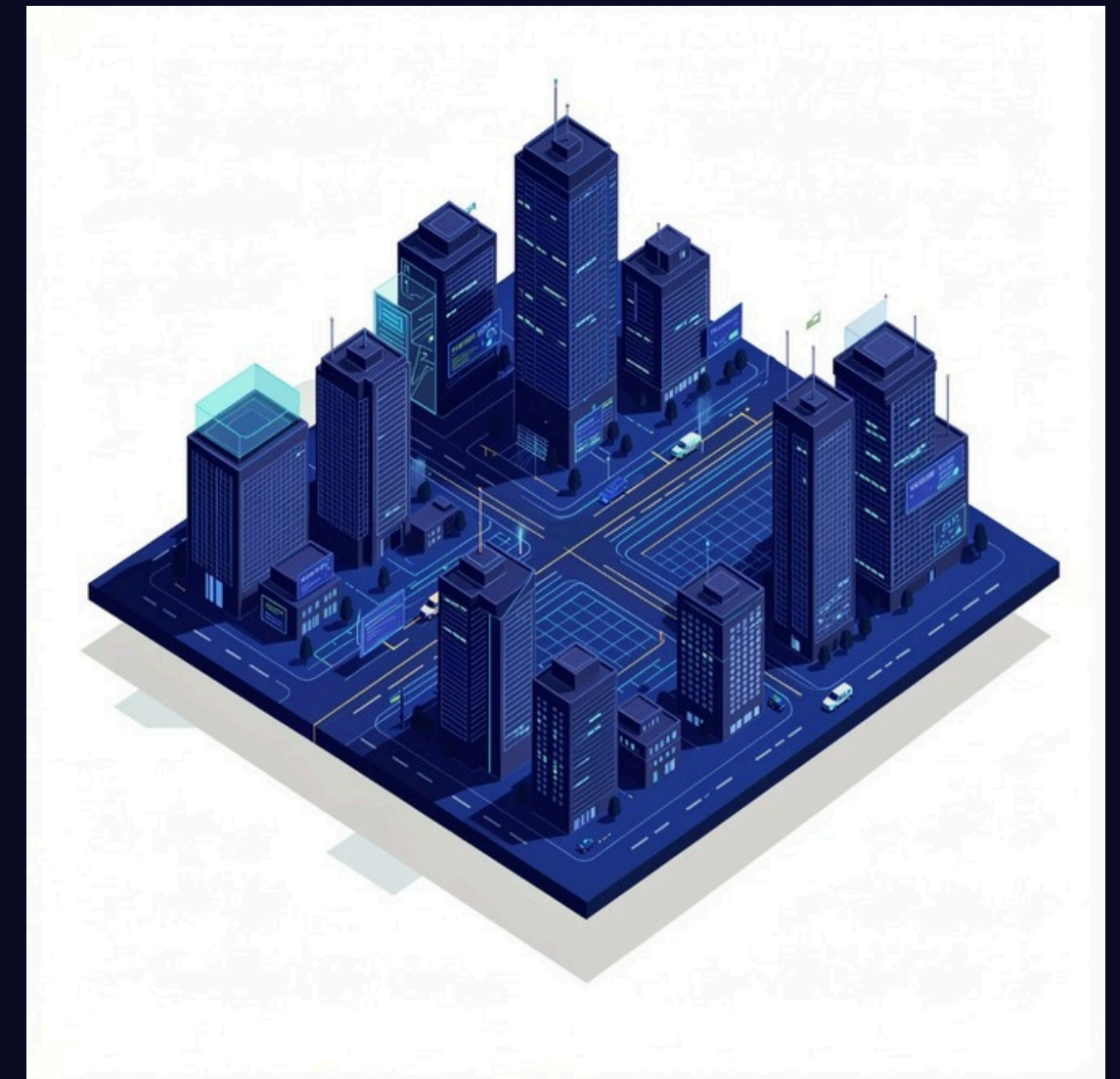
Decision-makers lack access to real-time insights, trends, or violation pattern data.



MOTIVATION

Why This System Matters Now

Urban traffic violations are rising at an alarming rate. Smart cities demand intelligent, automated systems that bring transparency, speed, and accountability to public infrastructure management.



Rising Violations

Traffic incidents have surged 40%+ in major cities over the last decade.

Smart City Push

Government initiatives mandate digital governance across urban services.

Automation Need

Manual processes can't scale — automation is the only viable path forward.

How the System is Structured

User Interface

React.js with Tailwind CSS for responsive UI

Middleware Layer

Logging, validation, auth, and error handling

Backend API

Node.js and Express.js exposing REST endpoints

Database

MongoDB for document storage and queries



Each layer communicates via RESTful APIs using Axios. Middleware intercepts every request for authentication, input validation, and centralized error handling — ensuring a robust and secure data pipeline.

Features Overview



Auth System

Secure Login & Signup with role-based access control for admins and officers.



Live Dashboard

Real-time statistics panel displaying violation counts, revenue, and trends.



Violation CRUD

Full Add, Edit, Delete, and Search capabilities with advanced filtering options.



Payment UI

Mock payment portal with multiple methods and auto-generated digital receipts.



Live Analytics

- Live analytics dashboard showing real-time updates of violations, fines, and trends

Smart Features That Set It Apart



Auto Fine Calculation

Fines computed instantly based on violation type, severity, and offense history.



Repeat Offender Detection

System flags drivers with 3+ violations and escalates fines automatically.



Number Plate Validation

Regex-based format validation ensures accurate plate number entry and lookup.



Data Analytics

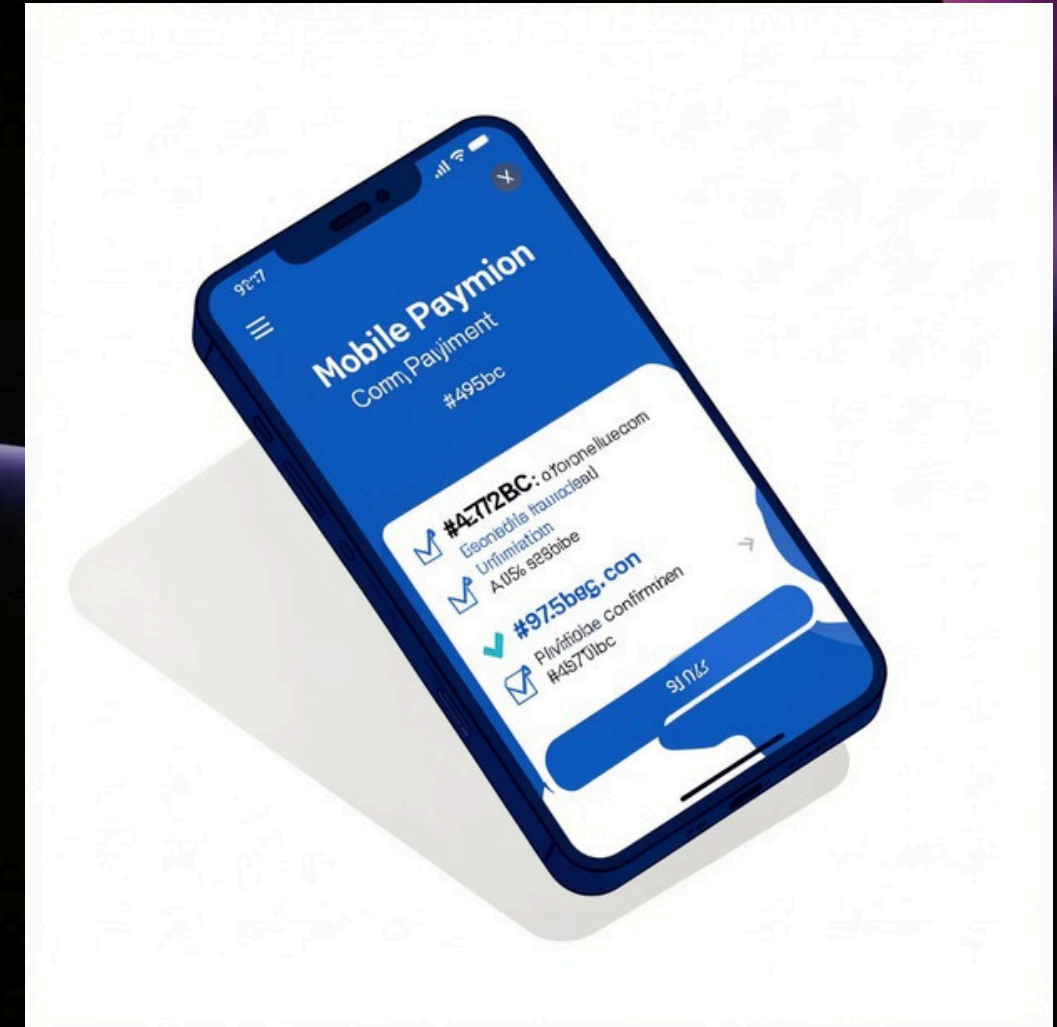
Interactive charts provide actionable insights into violation patterns over time.



PAYMENT SYSTEM

Flexible, Mock Payment Gateway

The system simulates a real-world payment experience with multiple methods — all without connecting to an actual payment processor. Upon completion, a digital receipt is auto-generated.



UPI

Credit / Debit Card

Net Banking

Cash



DASHBOARD

Data-Driven Decision Making



The analytics dashboard visualizes daily violation frequency and type distribution — empowering administrators to allocate enforcement resources where they matter most.

Technical Stack



Frontend

React.js + Tailwind

CSS — Component-driven UI with responsive, utility-first styling.



Backend

Node.js +

Express.js — Fast, scalable REST API with modular route architecture.



Database

MongoDB —

Flexible NoSQL document storage with Mongoose ODM schemas.



Libraries

Axios, Chart.js /

Recharts, Multer —

API calls, data viz, and file uploads.

WRAP-UP

Conclusion & Future Scope

This project successfully demonstrates a scalable, full-stack approach to modernizing traffic violation management — with real-world applicability in smart city deployments.

Future Roadmap



What We Built

A complete MERN-based system with CRUD, analytics, smart detection, and a payment UI.

Key Learnings

API integration, file uploads with Multer, Chart.js visualizations, and full-stack debugging.

1

Dispute Sysem

User can put disputes if they dont agree

2

Live Feed

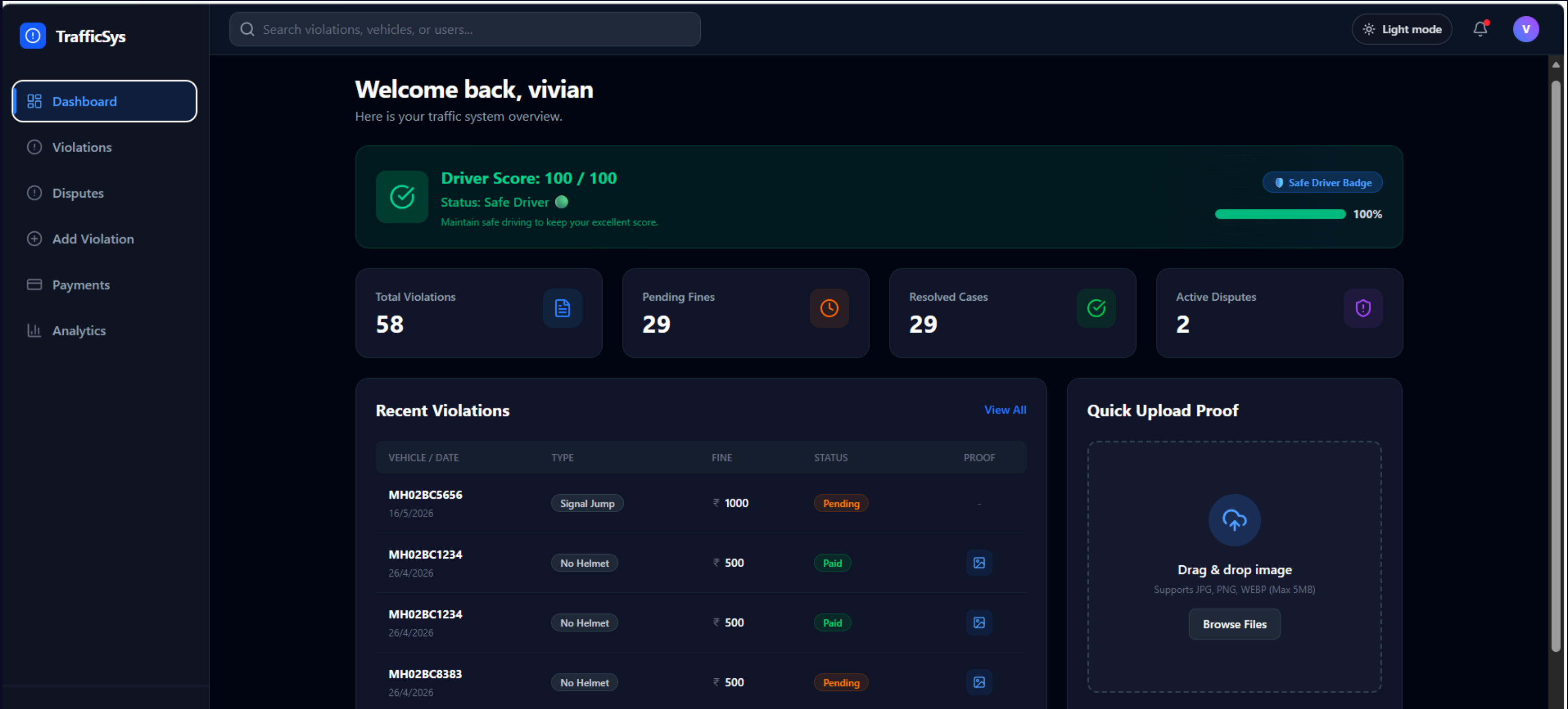
Real-time camera integration

3

Mobile App

Cross-platform iOS & Android version

Some Screenshots of the Application



ⓘ

TrafficSys

Dashboard

Violations

Disputes

Add Violation

Payments

Analytics

Sign Out

🔍

Search violations, vehicles, or users...

⚙️

Light mode

🔔

V

Violations

Manage and track traffic violations

+ Add Violation

🔍

Search by vehicle number or type...

📄

Export CSV

⌵

Filter

VEHICLE	TYPE	LOCATION	FINE	STATUS	DATE
MH02BC5656 Shlok Jain	Signal Jump	Borivali	₹ 1000	Pending	16/5/2026
MH02BC1234 Kshitij2	No Helmet	Bandra	₹ 500	Paid	26/4/2026
MH02BC1234 Kshitij2	No Helmet	Andheri	₹ 500	Paid	26/4/2026
MH02BC8383 Sam	No Helmet	Bandra	₹ 500	Pending	26/4/2026
Mh03VG8783 Niranjan	No Helmet	Bandra	₹ 500	Paid	25/4/2026
Mh03cv7383 Lakhan	No Helmet	Bhayandar	₹ 500	Paid	25/4/2026
MH03BV6838 Nitesh	No Helmet	Bhayandar	₹ 500	Pending	25/4/2026
MH02BC9229 Niraj	No Helmet	Bhayandar	₹ 500	Pending	25/4/2026
Mh02Bc8373	Over speed	Dadar	₹ 1500	Paid	25/4/2026

Vehicle Number

E.G. MH 01 AB 1234

Owner Name

Enter owner name

Vehicle Type

Two Wheeler

Violation Type

No Helmet

Auto-Detected Fine

Rs. 500 (Calculated automatically)

Location

Enter location

Proof of Violation (Image Upload)



Drag & drop image

Supports JPG, PNG, WEBP (Max 5MB)

Browse Files

Cancel

Submit Record

Pay Fines

Settle your pending traffic violation fines

Select Violation > 2 Payment Details > 3 Processing

Back

Select Payment Method

UPI

Credit / Debit Card

Net Banking

Card Number

1234 5678 9012 3456

Expiry (MM/YY)

08/27

CVV

...

Order Summary

Vehicle	MH02BC8383
Type	No Helmet
Fine	₹500
Convenience Fee	₹0

Total ₹500

Pay ₹500 Securely

256-bit Encrypted Mock Gateway



Disputes

Manage and review user disputes

USER INFO	VIOLATION	REASON	STATUS	DATE	ACTIONS
Kshitij2 Kshitij2@gmail.com	MH02BC1234 No Helmet	I was riding at the speed of 20 still officer fined me for no re...	Rejected	26/4/2026	
Kshitij2 Kshitij2@gmail.com	MH02BC1234 Overspeed	I was not overspeeding the officer without any reason fined ...	Rejected	26/4/2026	

System Analytics

Real-time overview of traffic violations and revenue

Total Revenue

Rs. 18,700



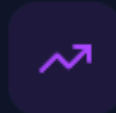
Total Violations

58



Recovery Rate

50%

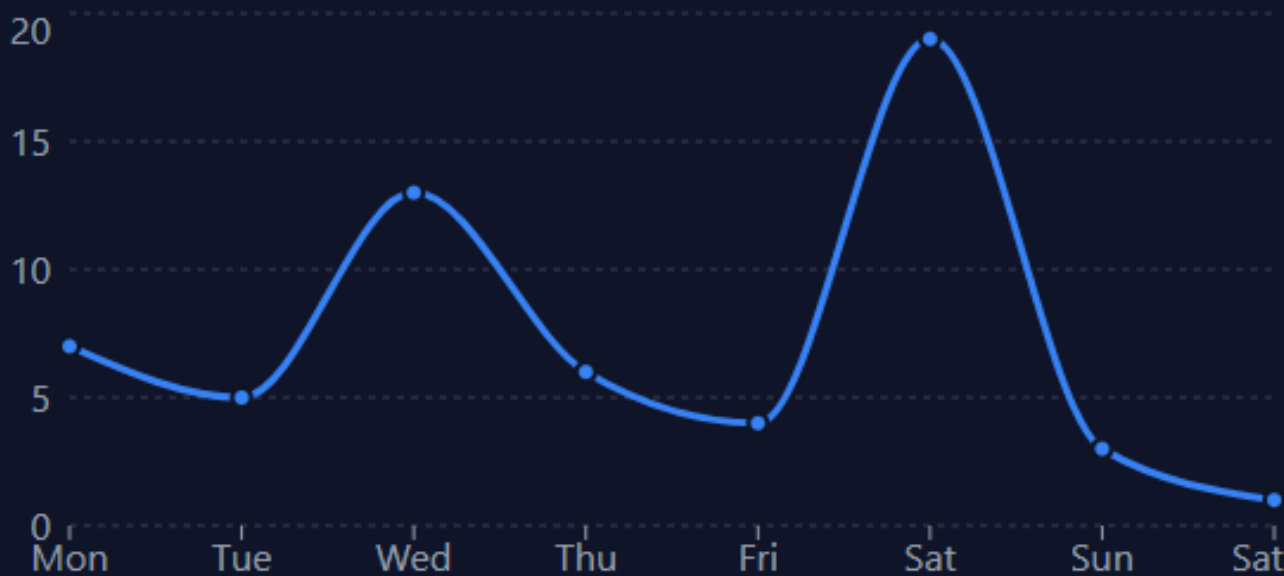


Repeat Offenders

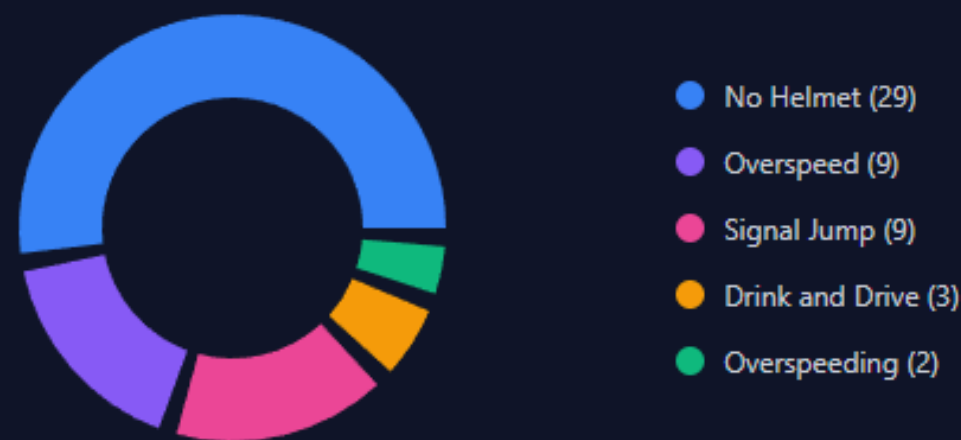
0



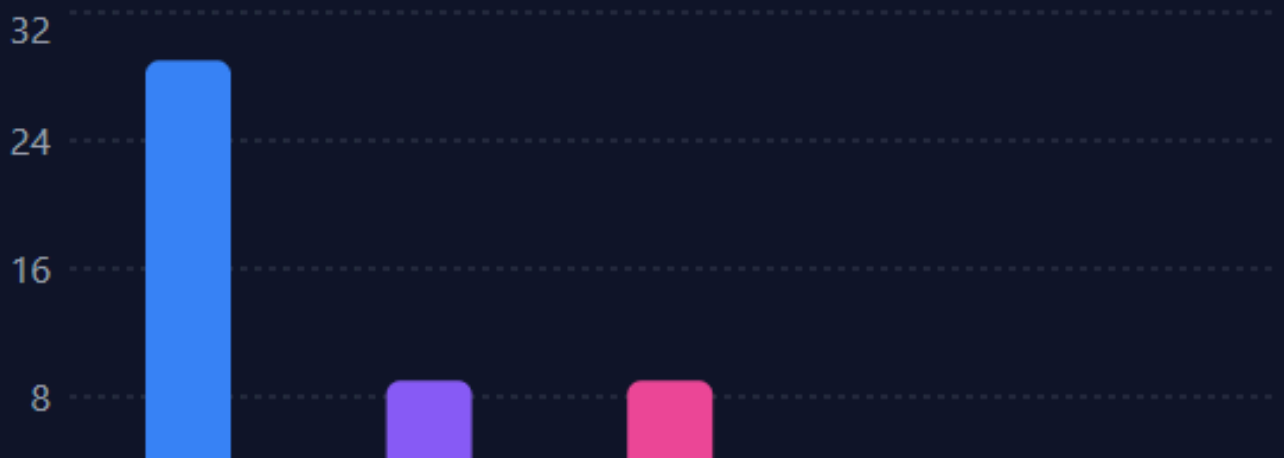
Daily Violation Trend (Line)



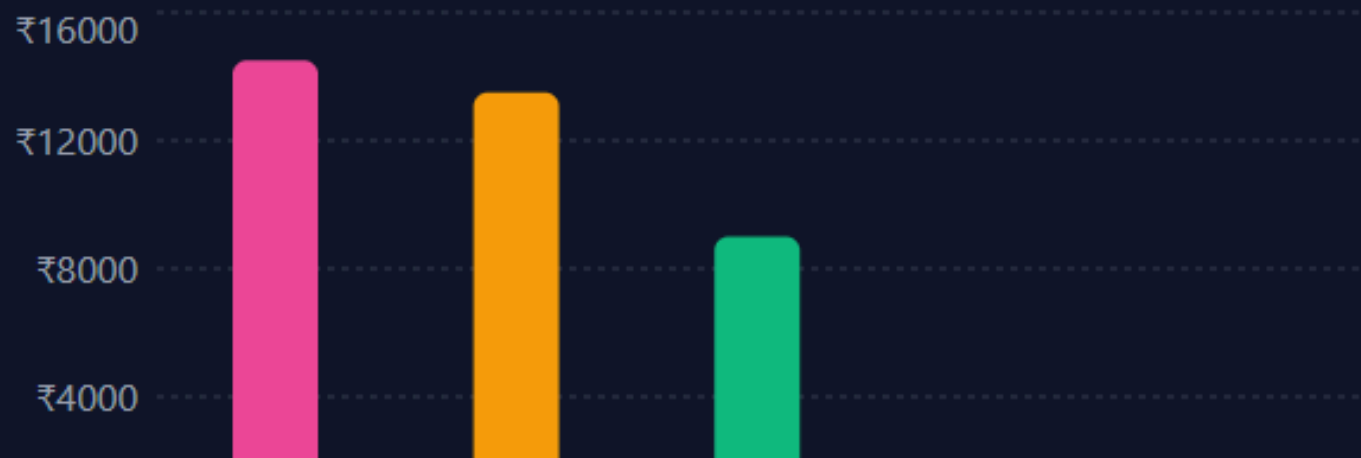
Violation Type Breakdown (Pie)



Violations by Type (Bar)



Revenue by Violation Type (Bar)



Thank You!

Vivan Shetty

10637

Cian Fonseca

10909

Soham More

10622

Computer Science Department
B.Tech Computer Engineering, Academic Year 2026